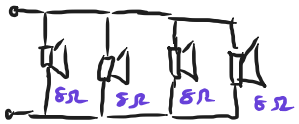
306. We have four loudspeakers of 8Ω and a mono amplifier of 100W at 8Ω with a minimum impedance of 2Ω.

a. Calculate the equivalent load impedance if we connect the four loudspeakers in parallel. (2,5 points)

With all 4 loudspeakers of 8Ω connected in parallel



For speakers, load impedance in parallel is $\frac{\text{impedance}}{\text{n° speakers}}$

As long as all speakers have the same Ω

$$Z_T = \frac{8\Omega}{4} = \boxed{2\Omega}$$

(in series, it would be the sum)

b. If a cluster formed by the four loudspeakers in parallel of the previous section generates 120dB SPL at 1m, calculate the theoretical level generated only by one loudspeaker at that distance. (2,5 points)

→ 120 dB SPL ↳ $dBV = 10^{(120/20)}$

→ 1m

Assuming coherent addition, everytime the signal is doubled, +6dB get added.

Thus, we substract -6dB -6dB for all 4 speakers, thus the theoretical level at that distance would be 120-6-6 = 108dB

Assuming incoherent addition though, which makes more sense for physically separated loudspeakers, we apply the same logic but with 3dB, thus

120-3-3 = **114dB**

Makes more sense since acoustic signals might differ, 114dB final answer

c. We are asked to add four additional loudspeakers connected also in parallel. Explain briefly if you would recommend this configuration. (2,5 points)

While 4 more loudspeakers in parallel will result in a total impedance of 8/8 = 1 Ohm, this might become an issue if the voltage is kept constant, as it can generate an unsafe current level.
In other words:

$V = RI$ — Increases / Const, Decreases

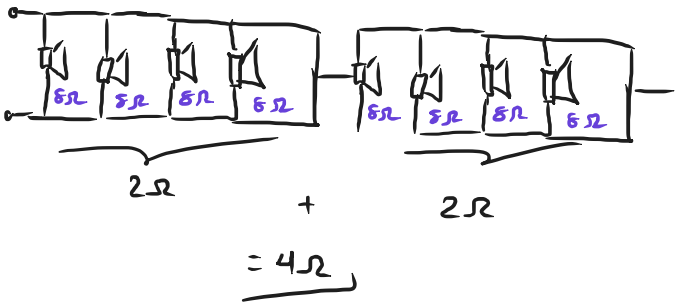
$P = VI$ — Increases / Constant, Increases

The reduction of impedance, by Ohm's law creates a noticeable increase of current, which can result in excessive heat inside the speaker, which might damage the device.. It can also affect the amplifier

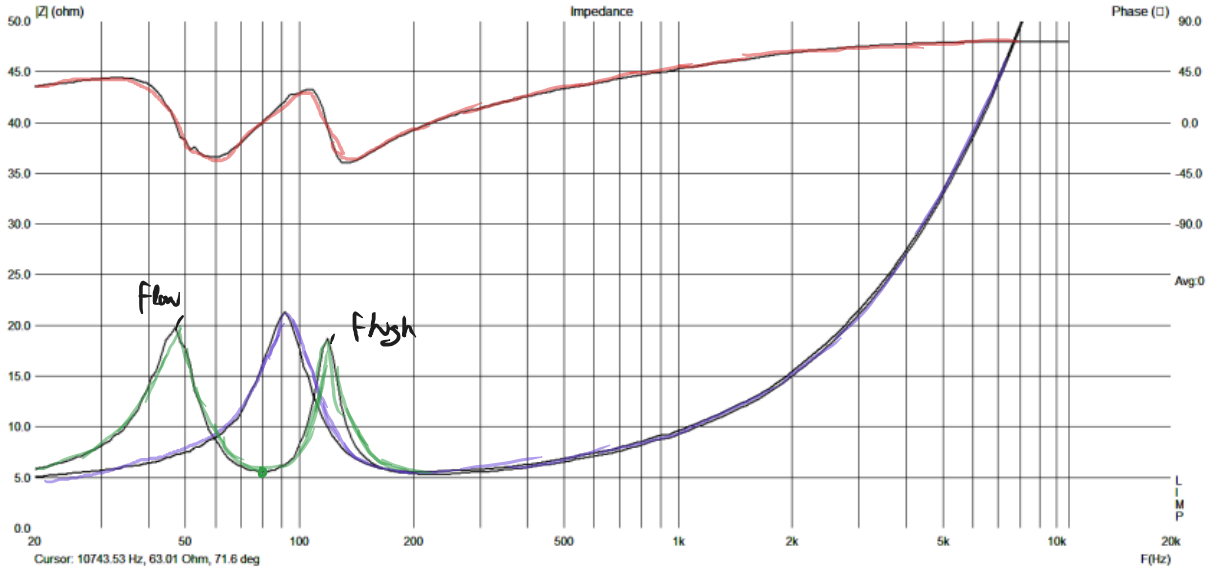
Additionally, we can observe that the amp requires a minimum impedance of 2 Ohms, thus we can't simply add 4 more in parallel

The smarter way to handle this would be placing these 4 speakers in another layout. We should deliver enough power but still using a safe amount of current

d. Illustrate through a diagram a solution to connect the eight loudspeakers to the amplifier with a total load impedance of 4Ω. (2,5 points)



307. For the next impedance response (magnitude and phase):



a. Justify which magnitude curve corresponds to the sealed box enclosure, and which one corresponds to the bass-reflex.

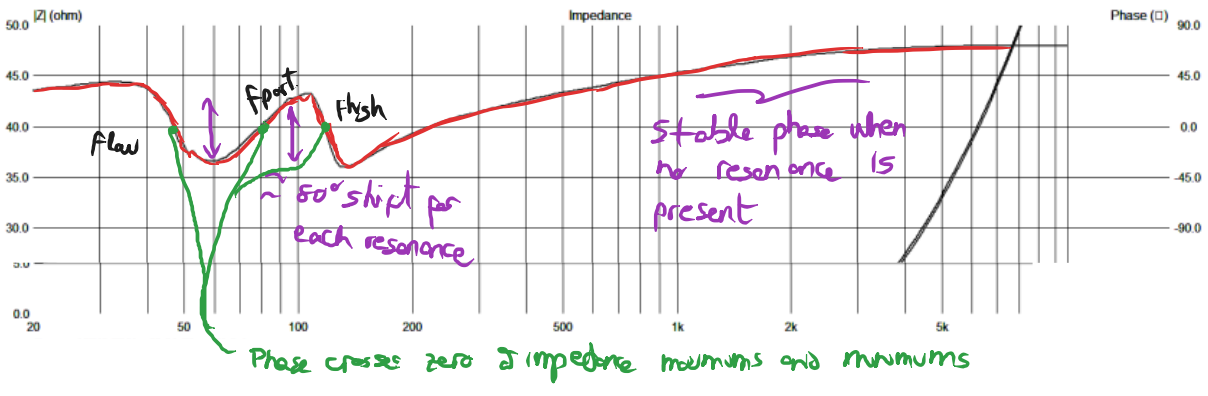
Bass reflex
- 2 peaks
- Lowest point between peaks
- flow < fhigh

Sealed box
- single peak with box resonance

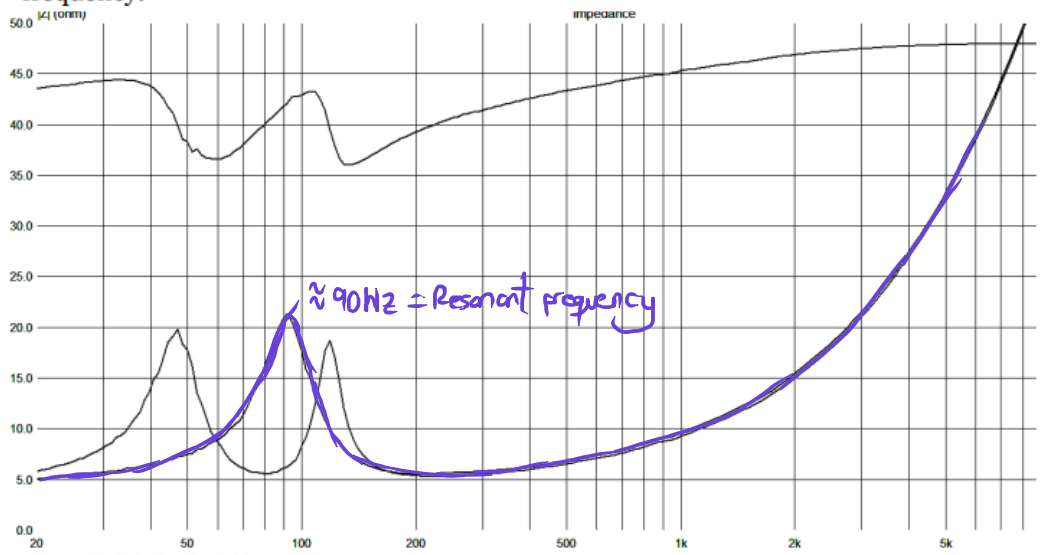
b. Justify briefly which of the two configurations is represented in the phase response.

By observing the phase curve (red), we can verify that it crosses zero 3 different times in a small range of frequencies, matching with the peaks and dips found in the bass reflex frequency response

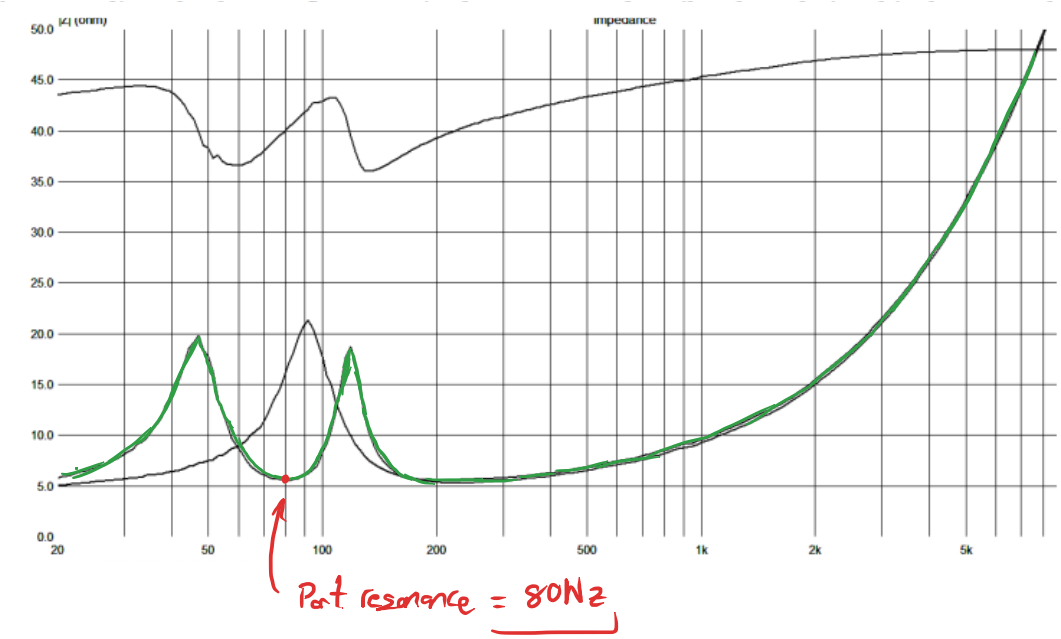
c. Indicate all the characteristic values that can be obtained from this phase response.



d. On the magnitude response of the sealed box, indicate the fundamental resonance frequency.



e. On the magnitude response of the bass-reflex configuration, indicate the resonance frequency of the port



f. Attending the bass-reflex magnitude response, describe the relationship between the resonant frequency of the box, and the resonant frequency of the driver.

Noticing that the first resonance peak (flow) has more impedance than the second (fhigh), we can infer that the resonance on the box (due to port) is higher than the resonant frequency of the driver.

Due to this mismatch, the bass reflex response might not provide the best low end. Nevertheless, the impedance difference between both peaks is small, so this "distorted" lower response won't be heard as much